Biometrika, 2025, pp. 1–5

doi: 10.1093/biomet/asaf075
Advance Access Publication Date: Day Month Year
Application Note

APPLICATION NOTE

lifelines-hc: a Python package for Higher Criticism testing in two-sample survival analysis under non-proportional hazards

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FOR PUBLISHER ONLY Received on Month ; revised on Month ; accepted on Month

Abstract

Motivation: The log-rank test is a standard tool for two-sample survival comparison and has good power against the widely studied proportional-hazards alternatives. However, Kipnis et al. [2026] showed that it can lose power against sparse hazard departures, where the hazard difference is concentrated in a small number of time intervals whose locations along the follow-up are unknown a priori. Weighted log-rank tests, such as Gehan-Wilcoxon, Tarone-Ware, Peto-Prentice, and Fleming-Harrington, each impose a pre-specified temporal emphasis, and are therefore also insensitive to sparse departures occurring outside their emphasized time regions. **Results:** We introduce `lifelines-hc`, a Python package extending the `lifelines` survival library with the HCHG test: Higher Criticism (HC) applied to per-interval HyperGeometric (HG) p-values from Kipnis et al. [2026]. HCHG is a test for right-censored two-sample survival data. It detects sparse hazard departures at unknown locations without committing to any temporal pattern. Across four biological case studies — an immuno-oncology trial (CheckMate 057 PFS, $n = 582$), a targeted kinase-inhibitor trial in metastatic prostate cancer (COMET-1 OS, $n = 1028$), an adjuvant bisphosphonate trial (AZURE DFS, $n = 3359$), and the Copenhagen Study group for Liver diseases (CSL) liver-cirrhosis trial with fully public individual patient data ($n = 446$) — HCHG achieves $p \leq 0.014$ in every dataset, while the log-rank test is non-significant throughout ($p \geq 0.26$); among the weighted log-rank tests, only Gehan-Wilcoxon and Peto-Prentice on CheckMate 057 reach $p < 0.05$.
Availability and Implementation: `lifelines-hc` is freely available under the MIT licence at `https://github.com/TODO/lifelines-hc` and via `pip install lifelines-hc`; it requires Python ≥ 3.8 and `lifelines` ≥ 0.29. Full analysis scripts and per-method results tables are provided as Supplementary Data.

Key words: survival analysis; higher criticism; sparse signal detection; hypergeometric test; hypothesis testing; Python

Introduction

The Kaplan-Meier estimator [Kaplan and Meier, 1958] and the log-rank test [Mantel, 1966] are the workhorses of two-sample survival analysis in biomedical research. The log-rank statistic has good power against a wide range of useful alternatives, particularly proportional-hazards (PH) alternatives [Schoenfeld, 1981]. Nevertheless, Kipnis et al. [2026] showed that the log-rank test is not optimal against *sparse* hazard departures, in which the hazard difference is concentrated in a small, unknown subset of time intervals.

Such localized or non-proportional departures arise naturally in several biomedical settings. Immune checkpoint inhibitors may require an immune-priming phase before producing durable tumour suppression, leading to delayed treatment effects and crossing survival curves; for example, in the second-line nonsquamous NSCLC trial CheckMate 057, nivolumab improved overall survival relative to docetaxel, while the progression-free-survival curves crossed, consistent with delayed benefit [Borghaei et al., 2015]. Targeted therapies can also produce endpoint-specific benefits over a limited follow-up window that are diluted in global survival summaries: in the COMET-1 mCRPC trial, cabozantinib markedly improved bone-scan response at week 12 and radiographic progression-free survival, but did not significantly improve overall survival [Smith et al., 2016]. Subgroup-dependent biology can similarly generate composite, time-varying hazard differences: in the AZURE trial of adjuvant zoledronic acid, no overall benefit was observed in the full population, but benefit was seen among women with established menopause, consistent with an effect mediated by the postmenopausal bone microenvironment [Coleman et al., 2011, 2014]. Time-varying treatment effects can likewise produce multi-window hazard departures: in the CSL liver-cirrhosis trial, prednisone was harmful early (infection and fluid retention) but beneficial later through anti-inflammatory action, yielding survival curves that cross more than once [Martinussen and Scheike, 2006]. This canonical non-proportional-hazards dataset is widely used to illustrate time-varying-coefficient models and provides fully public individual patient data via the CRAN `timereg` package [Martinussen and Scheike, 2006] and the SurvSet repository [Drysdale, 2022].

The need for log-rank alternatives in survival analysis is well recognised, and several weighted log-rank tests have been developed for non-proportional-hazards settings [Gehan, 1965, Peto and Peto, 1972, Tarone and Ware, 1977, Fleming and Harrington, 1991]. Each applies a temporal weight function, emphasising early, intermediate, or late events. Each of these tests achieves good power when the signal matches the intended temporal pattern. However, choosing an appropriate weight requires knowing in advance *where* the hazard departure will occur. When the departure is sparse but its location is unknown a priori, all weighted log-rank tests can miss it simultaneously, as we demonstrate below.

Kipnis et al. [2026] proposed the HCHG test — the Higher Criticism (HC) statistic of Donoho and Jin [2004] applied to per-interval HyperGeometric (HG) p-values — specifically for this setting. HCHG divides the follow-up into equal-width intervals, derives a hypergeometric p-value for the event-count comparison in each interval under the null of equal group-specific hazards, and applies the HC statistic to test whether the body of these p-values is collectively implausible under the global null. Because HCHG does not pre-specify any temporal weight, it retains power for sparse departures wherever they occur along the follow-up axis.

We present `lifelines-hc`, a Python package that implements HCHG and integrates with the popular `lifelines` library [Davidson-Pilon, 2019].

The `lifelines-hc` package

The HCHG statistic

HCHG [Kipnis et al., 2026] combines two components: hypergeometric interval p-values and the HC statistic of Donoho and Jin [2004, 2015].

Hypergeometric interval p-values. Given two groups with right-censored observations (T_i, E_i, G_i), partition the follow-up range into K equal-width intervals. For interval k, let n_k be the total events, r_k^A and r_k^B the subjects at risk in groups A and B, and x_k the events in group B. Under the null of equal group-specific hazards,

$$x_k \,\Big|\, n_k,\, r_k^A,\, r_k^B \;\sim\; \text{Hypergeometric}\Big(r_k^A + r_k^B,\; r_k^B,\; n_k\Big). \quad (1)$$

yielding a one-sided p-value p_k for each interval using an exact test.

HC aggregation. Ordering the K p-values as $p_{(1)} \le \cdots \le p_{(K)}$, the HC statistic [Donoho and Jin, 2004, 2015] is applied to this collection:

$$\text{HCHG}_K \;=\; \max_{1\le j\le \lfloor \gamma K\rfloor} \frac{j/K - p_{(j)}}{\sqrt{p_{(j)}\big(1-p_{(j)}\big)/K}}, \quad (2)$$

where $\gamma \in (0,1)$ (default 0.2) restricts scanning to the most extreme fraction of interval p-values [Donoho and Jin, 2004, 2015]. A global p-value is obtained by permuting group labels ($n_{\text{perms}} = 2000$ in the analyses reported here). For the two-sided alternative, the minimum of the two one-sided interval p-values replaces p_k in (2).

Software design and API

`lifelines-hc` exposes a function-based API consistent with `lifelines.statistics`, accepting NumPy arrays of event times and indicators:

```
from lifelines_hc import (higher_criticism_test,
                          fisher_combination_test
                              ,
                          min_p_test,
                          suspected_deviations)

result = higher_criticism_test(
    T_A, T_B,
    event_observed_A=E_A, event_observed_B=E_B,
    n_intervals_to_pool=100, n_permutations
        =2000,
    alternative='both', gamma=0.2, seed=42)

print(result.p_value)          # permutation-
    calibrated p-value
print(result.test_statistic)  # HCHG statistic
    value
```

The companion `suspected_deviations()` function returns a `pandas.DataFrame` of per-interval p-values with a `suspected` boolean column marking HCHG-flagged intervals, enabling overlay of diagnostic shading on Kaplan-Meier plots. Two complementary omnibus tests are also provided: Fisher combination [Fisher, 1932], which aggregates evidence via $-2\sum_k \log p_k$, and a Bonferroni-corrected MinP test. All three share the same permutation infrastructure and the same hypergeometric interval p-values. The package additionally ships with plotting utilities (`plot_km_with_hc`, `plot_pvalue_profile`) and a convenience wrapper `run_all_tests()` that benchmarks HCHG, Fisher, MinP, log-rank, and the four weighted alternatives in a single call, returning a `pandas.DataFrame` for easy comparison.

Results

We demonstrate `lifelines-hc` on four biological datasets with distinct non proportional-hazards (PH) structures (Figure 1), benchmarking HCHG against the log-rank test and four weighted alternatives. Individual patient data for the first three trials were reconstructed from published Kaplan-Meier curves; the CSL trial provides publicly available individual patient data. Full per-method p-value tables are provided in Supplementary Data.

Immuno-oncology: crossing survival curves

CheckMate 057 [Borghaei et al., 2015] randomised 582 patients with advanced non-squamous NSCLC to nivolumab ($n = 292$) or docetaxel ($n = 290$). Progression-free survival showed crossing curves: docetaxel provided a short-term advantage during immune priming, subsequently overtaken by the durable nivolumab response (Figure 1A). Individual patient data were reconstructed from the published Kaplan-Meier curve via the algorithm of Guyot et al. [2012] as implemented in the `kmdata` R package.

The log-rank test is non-significant ($p = 0.351$) as the early excess and late benefit partially cancel. Among the weighted alternatives, Gehan-Wilcoxon ($p = 0.041$) and Peto-Prentice ($p = 0.049$) reach borderline significance by detecting the early crossing, but Tarone-Ware ($p = 0.358$) and Fleming-Harrington (1,1) ($p = 0.256$) do not. HCHG achieves $p = 0.001$ and characterises the full crossing structure with far greater power (Figure 1A,E) by flagging both the early divergence and late separation windows.

Targeted therapy in prostate cancer: a transient early benefit

The COMET-1 trial [Smith et al., 2016] randomised 1028 patients with chemotherapy-pretreated mCRPC to cabozantinib ($n = 682$) or prednisone ($n = 346$) in a 2:1 ratio. Overall survival showed no significant log-rank difference ($p = 0.262$). Cabozantinib, by inhibiting MET and VEGFR2 kinases, significantly improved bone scan response at week 12 (a pre-specified secondary endpoint), confirming genuine biological activity against bone-metastatic disease. This short-term activity did not translate into sustained overall survival (OS) benefit, however, as resistance through alternative survival pathways rapidly emerged. The result is a temporally concentrated early OS advantage (months 3–7) that is subsequently lost (Figure 1B,F).

All four weighted alternatives are non-significant ($p \geq 0.19$): the early-weighted Gehan-Wilcoxon ($p = 0.193$) and Peto-Prentice ($p = 0.206$) come closest but are diluted by the null period after the bone-response window. HCHG ($p = 0.014$) aggregates evidence from the specific early intervals where the transient effect is concentrated.

Adjuvant bisphosphonate therapy: a menopause-dependent effect

The AZURE trial [Coleman et al., 2011, 2014] randomised 3359 early-stage breast cancer patients to standard adjuvant therapy with ($n = 1681$) or without ($n = 1678$) zoledronic acid. Disease-free survival showed no significant log-rank difference ($p = 0.305$) because the drug's bone-microenvironment benefit depends on the low-oestrogen state of postmenopause, which was absent in the many premenopausal patients enrolled at baseline. As these patients progressively transitioned to postmenopause during the decade-long follow-up, the hazard difference accumulated in a specific mid-to-late window (months 20–60) rather than uniformly (Figure 1C,G).

All four weighted alternatives are non-significant ($p \geq 0.28$): no single temporal weight captures a pattern distributed across this window while remaining unaffected by the null early period. HCHG ($p = 0.005$) detects this scattered but collectively significant signal. The result is validated by published subgroup analyses showing significant disease-free survival (DFS) benefit in postmenopausal patients [Coleman et al., 2011, 2014], confirming that HCHG is detecting a genuine biological effect.

Corticosteroid therapy in liver cirrhosis: a time-varying effect

The CSL trial [Martinussen and Scheike, 2006] randomised 446 patients with histologically verified liver cirrhosis to prednisone ($n = 226$) or placebo ($n = 220$) in Copenhagen (1962–69), with 270 deaths over follow-up of up to $\sim$13 years. Unlike the preceding case studies, the raw individual patient data are genuinely public—distributed in the CRAN `timereg` package (dataset `csl`) and the open SurvSet repository [Drysdale, 2022]—rather than reconstructed from a published Kaplan-Meier curve.

Corticosteroids exert a time-varying effect in cirrhosis: early harm through immunosuppression, infection, and fluid retention is followed by later anti-inflammatory benefit in actively inflamed subgroups. The survival curves cross more than once, with the treatment effect concentrated in a few non-contiguous windows (Figure 1D,H).

The log-rank test is non-significant ($p = 0.384$), and all four weighted alternatives are as well ($p \geq 0.13$): Gehan-Wilcoxon ($p = 0.51$), Tarone-Ware ($p = 0.38$), Peto-Prentice ($p = 0.45$), and Fleming-Harrington (1,1) ($p = 0.14$). No single temporal weight captures a multi-window, sign-changing departure. HCHG achieves $p = 0.002$ and Fisher combination $p = 0.034$, demonstrating detection of sparse non-proportional-hazards departures on genuinely public raw data.

Across the four case studies, HCHG achieves $p \leq 0.014$ in every domain, while the log-rank test is non-significant throughout ($p \geq 0.26$). The two borderline significances for Gehan-Wilcoxon ($p = 0.041$) and Peto-Prentice ($p = 0.049$) in CheckMate 057 are the only instances where any weighted alternative reaches $p < 0.05$; elsewhere the weighted tests return $p \geq 0.13$. These reflect their early emphasis coincidentally aligning with the crossing-curve signal rather than general sensitivity to non-PH. The four patterns differ—early and late crossing, transient early benefit, mid-to-late accumulated effect, and multi-window sign-changing departure—yet HCHG detects all without per-dataset tuning. Fisher combination is also significant in all four domains, consistent with sparse localised signals.

Conclusion

`lifelines-hc` provides an accessible Python implementation of HCHG for two-sample right-censored survival data. HCHG is specifically designed for sparse hazard departures at unknown temporal locations: unlike weighted log-rank alternatives, it requires no a priori specification of where the hazard difference will occur and retains power for any concentrated departure along the follow-up axis. The package integrates transparently with `lifelines` workflows, returns diagnostic interval-level information identifying which time windows drive the signal, and includes Fisher combination and MinP tests as complementary omnibus alternatives sharing the same hypergeometric infrastructure.

Conflict of Interest

None declared.

Funding

The work of Alon Kipnis was supported in parts by the US-Israel Binational Science Foundation (BSF) grant 2022124.

Data availability

Individual patient data for CheckMate 057, COMET-1, and AZURE were reconstructed from published Kaplan-Meier curves using the algorithm of Guyot et al. [2012] as implemented in the `kmdata` R package (`https://github.com/raredd/kmdata`). CSL liver-cirrhosis individual patient data are fully public via the CRAN `timereg` package (dataset `csl`) and the SurvSet repository [Drysdale, 2022] (`https://github.com/ErikinBC/SurvSet`). All analysis code is available at `https://github.com/TODO/lifelines-hc/tree/main/AppNote`.

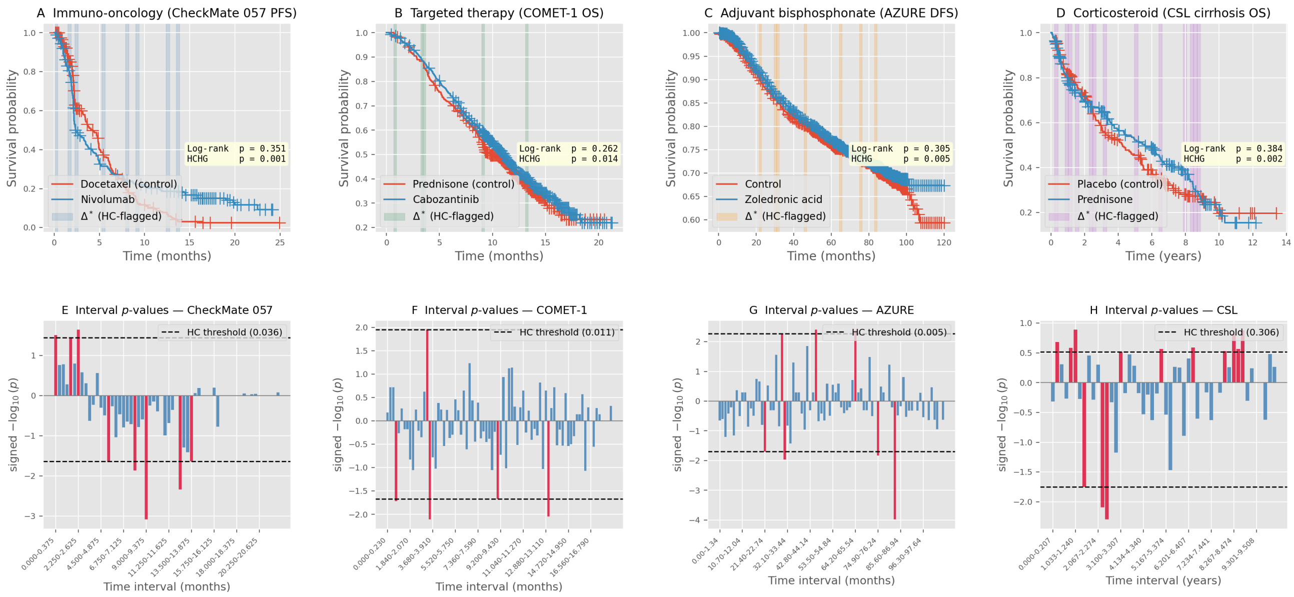

Fig. 1. Higher Criticism detects non-proportional hazard departures across four clinical datasets with distinct temporal patterns. Top row (A–D): Kaplan-Meier curves with HCHG-flagged intervals (shaded). Bottom row (E–H): per-interval *signed* $-\log_{10}$(hypergeometric p-value). Upward bars mark excess events in the treatment arm, downward bars excess in the control arm. Dashed lines are the per-direction HCHG thresholds; bars reaching beyond either line are highlighted in red and coincide exactly with the intervals shaded above. **(A,E)** CheckMate 057 PFS (nivolumab vs. docetaxel, $n = 582$); log-rank $p = 0.351$, HCHG $p = 0.001$. Curves cross at $\sim$3 months before nivolumab gains a durable late advantage. **(B,F)** COMET-1 OS (cabozantinib vs. prednisone, $n = 1028$); log-rank $p = 0.262$, HCHG $p = 0.014$. Early OS advantage (months 3–7) during bone-lesion response, fading as resistance develops. **(C,G)** AZURE DFS (zoledronic acid vs. control, $n = 3359$); log-rank $p = 0.305$, HCHG $p = 0.005$. Menopause-dependent benefit accumulates in months 20–60. **(D,H)** CSL OS (prednisone vs. placebo, liver cirrhosis, $n = 446$); log-rank $p = 0.384$, HCHG $p = 0.002$. Time-varying corticosteroid effect with multiple curve crossings; *fully public* raw individual patient data (CRAN `timereg` / SurvSet). In every dataset the log-rank test is non-significant while HCHG is significant. For the first three, individual patient data reconstructed via Guyot et al. [2012].

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